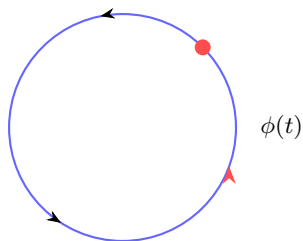


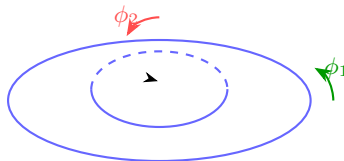
From Oscillatory Coupling to Toroidal Neural Manifolds

(A) Single Oscillator



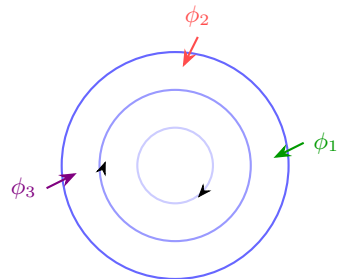
State space: S^1

(B) Two Coupled Oscillators



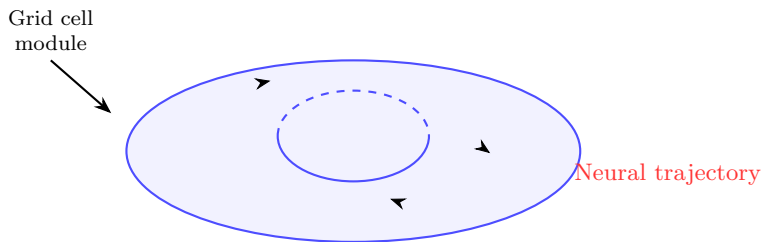
State space: $T^2 = S^1 \times S^1$

(C) Three Coupled Oscillators



State space: $T^3 \subset S^3$
Quaternion representation

(D) Grid Cell Population Activity on T^2



Gardner et al. (2022): persistent cohomology
reveals T^2 topology in grid cell ensembles